CS 391
Web Application Development

Esma Meral

esma.meral@ozyegin.edu.tr

Week - 3

Outline

1. Responsive Web Design
2. Media Queries
3. What is Bootstrap?
4. Get Started with Bootstrap
5. Layout
6. Content & Components

Responsive Web Design

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What is Responsive Web Design?

- A responsive web design will automatically adjust for different screen sizes and viewports.
- Web pages should not leave out information to fit smaller devices, but rather adapt its content to fit any device:

What is Responsive Web Design?

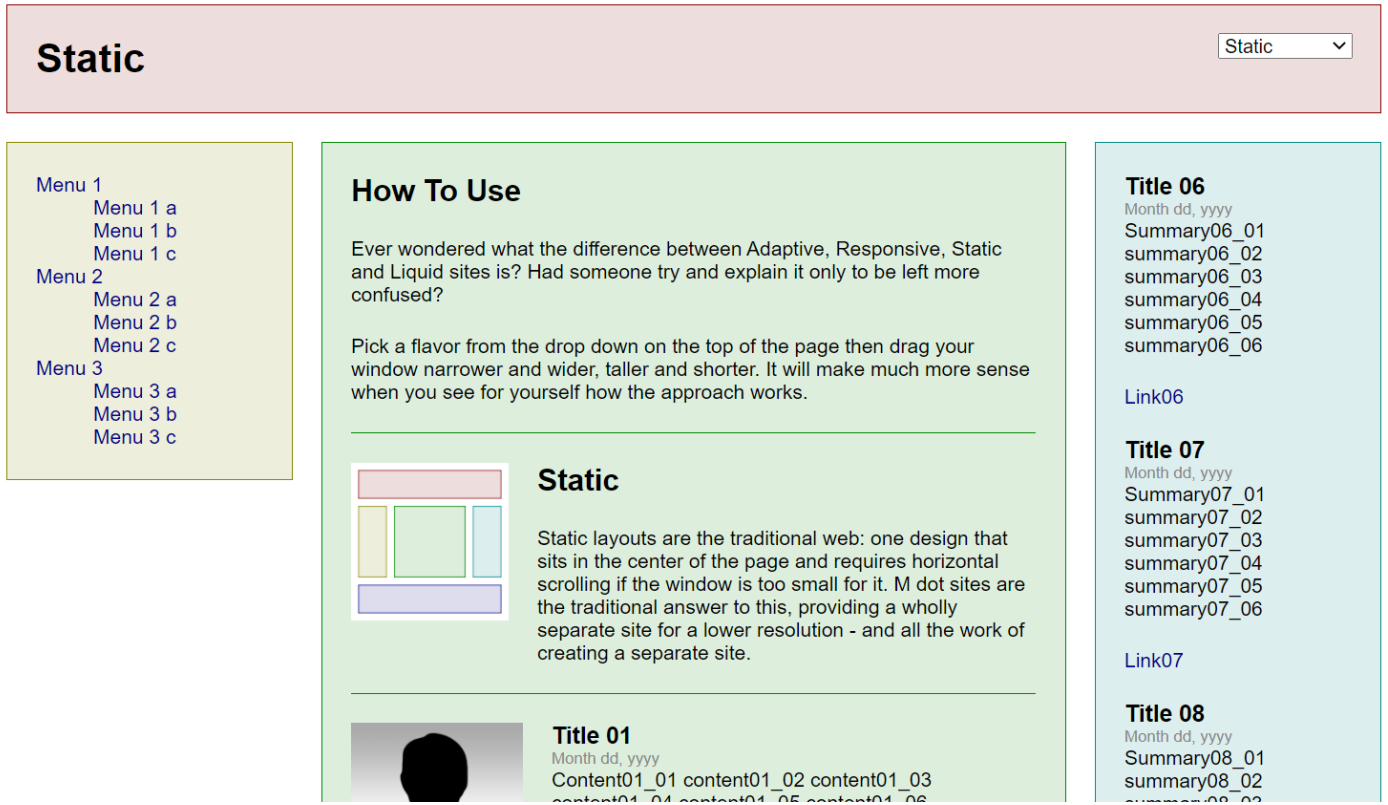
- Responsive web design is not a program or a JavaScript
- Responsive web design uses only HTML and CSS

Web Design Approaches

- Fixed(static)
- Fluid(liquid)
- Adaptive
- Responsive

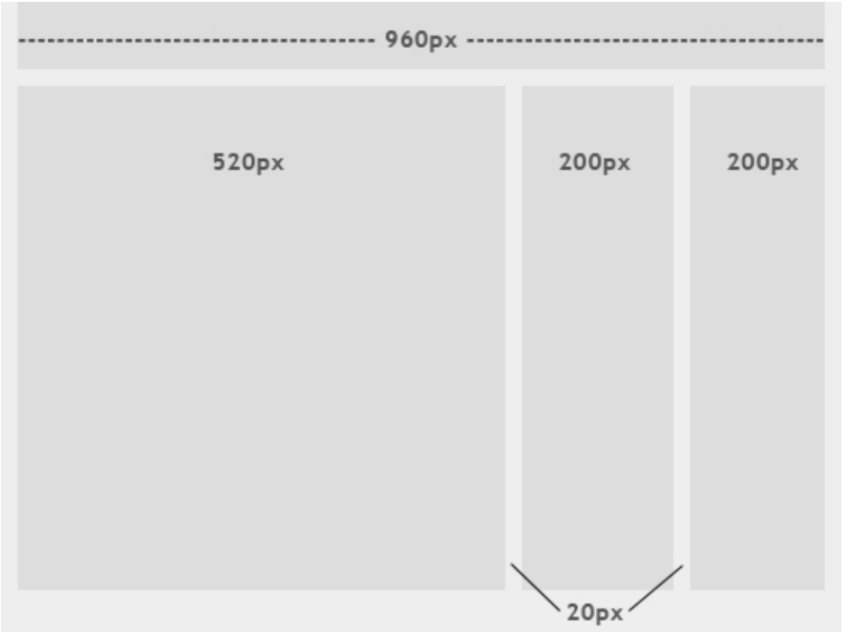
Difference between Adaptive, Responsive, Static and Liquid sites?

- http://g-mops.net/epica_saitama/epica_layout/index_adaptive.html



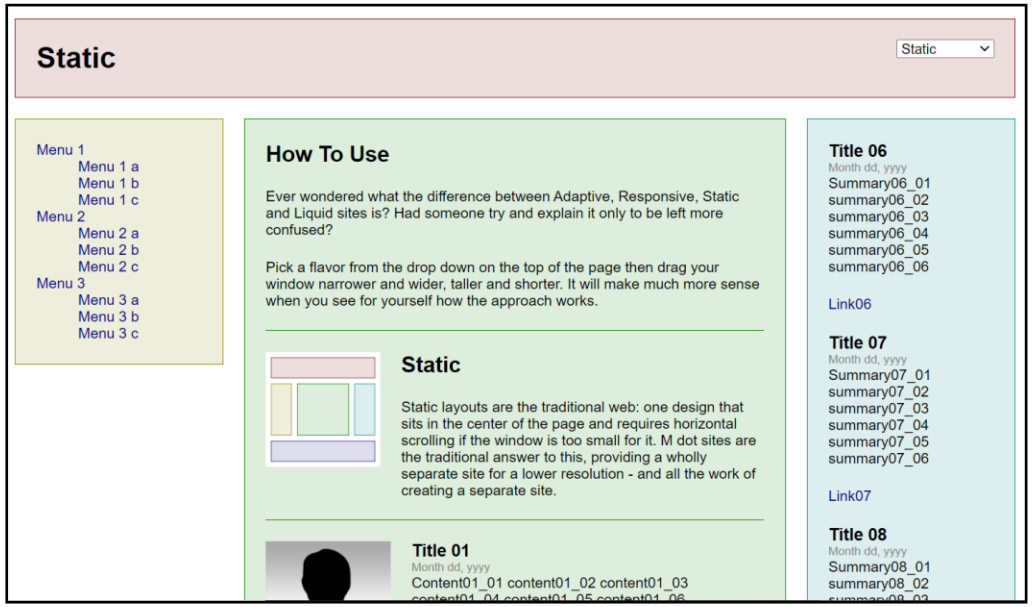
Fixed (static)…

- **Fixed** websites **<u>have a set width</u>** and resizing the browser or viewing it on different devices won't affect on the way the website looks.

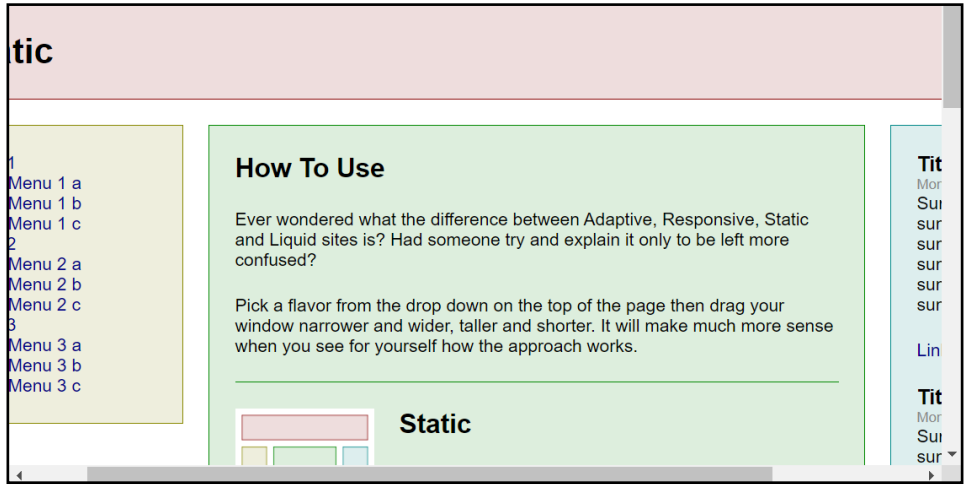


...Fixed(static)

Large screen



Small screen



Vertical scroll bar

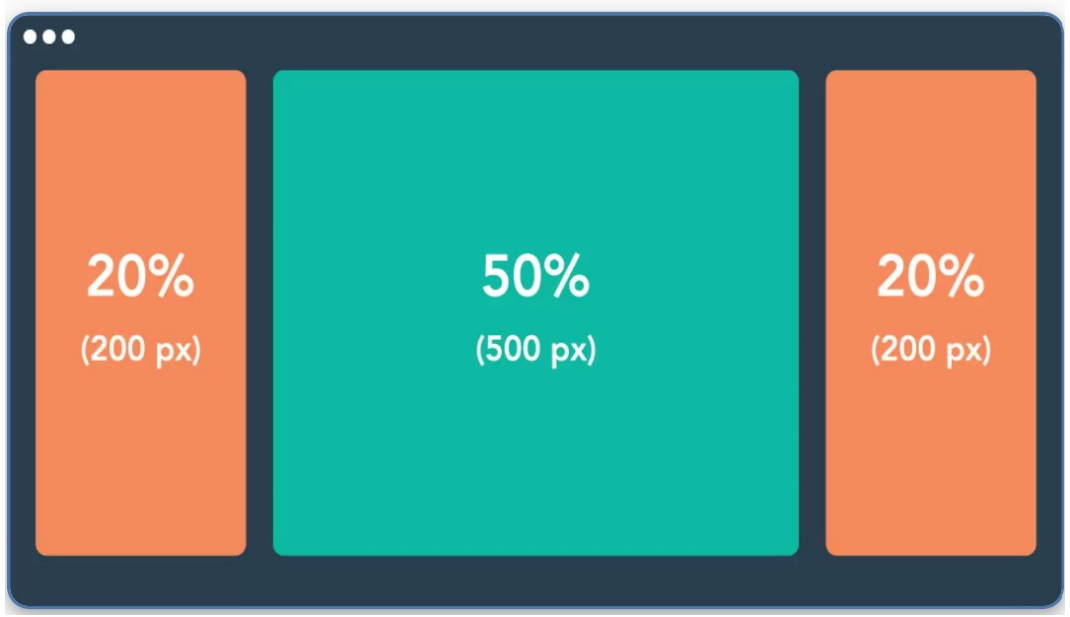
Horizontal scroll bar

Fluid(liquid)…

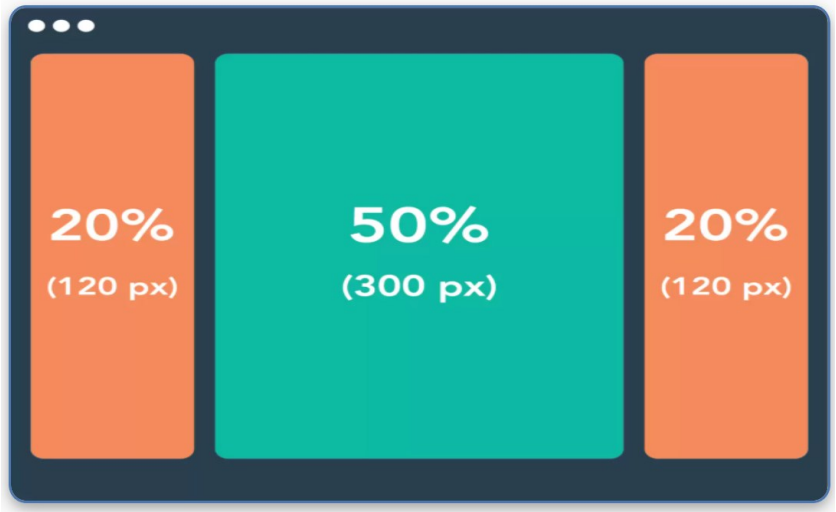
- **Fluid** websites are built using percentages for widths. As a result, columns are **relative** to one another and the browser allowing it to **scale up and down fluidly**.
- Fluid layout adapts itself to different browser window sizes : all the values used are calculated proportionally to the viewport size, so when resizing, all the columns are resized, **but the general layout stays the same**.

...Fluid(liquid)

1000 pixel wide browser window



600 pixel wide browser window



When we minimize the browser window to just 600px, a fluid design scales these elements proportionally

general layout stays the same.

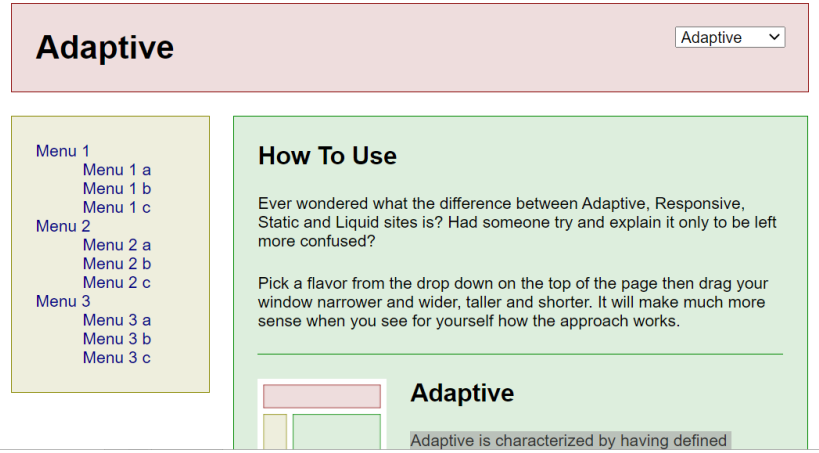
Adaptive

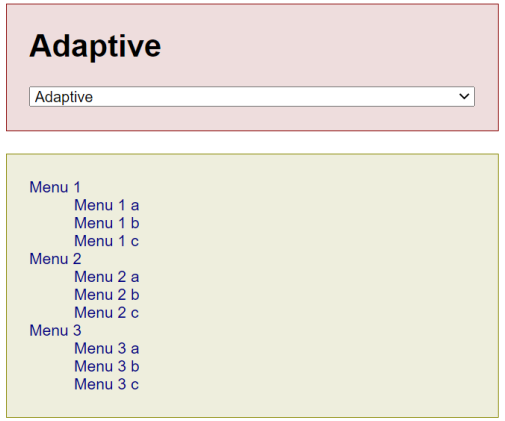
- **Adaptive** websites introduce **media queries** to target specific device sizes, like smaller monitors, tablets, and mobile.
- Adaptive is characterized by having defined **layouts for different resolutions.** Within each layout, resizing the window does not change the layout.
- You can think of Adaptive as a **series of static layouts**.

Adaptive

Large screen : 3 columns layout

Small screen: 2 columns layout

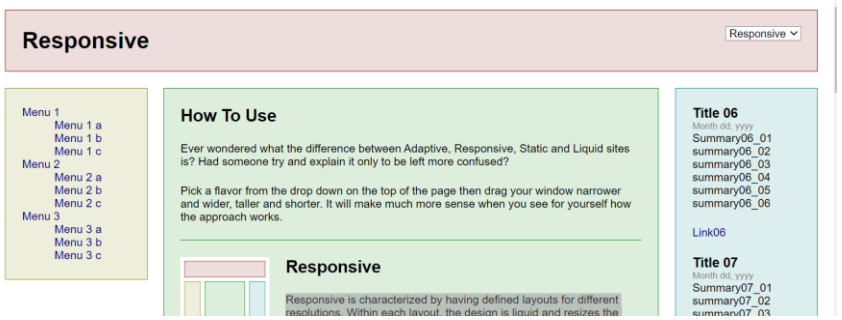




Smallest screen: 1 column layout

Responsive

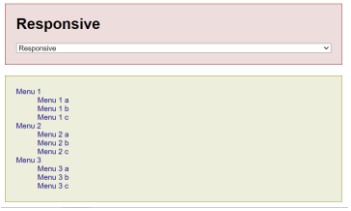
- Responsive is characterized by **having defined layouts for different resolutions**. Within each layout, the **design is liquid and resizes** the width of elements relative to the changing window size.
- You can think of Responsive as a **series of Liquid layouts**.



Large screen :
3 columns layout

Smaller screen :
3 columns layout, narrow columns

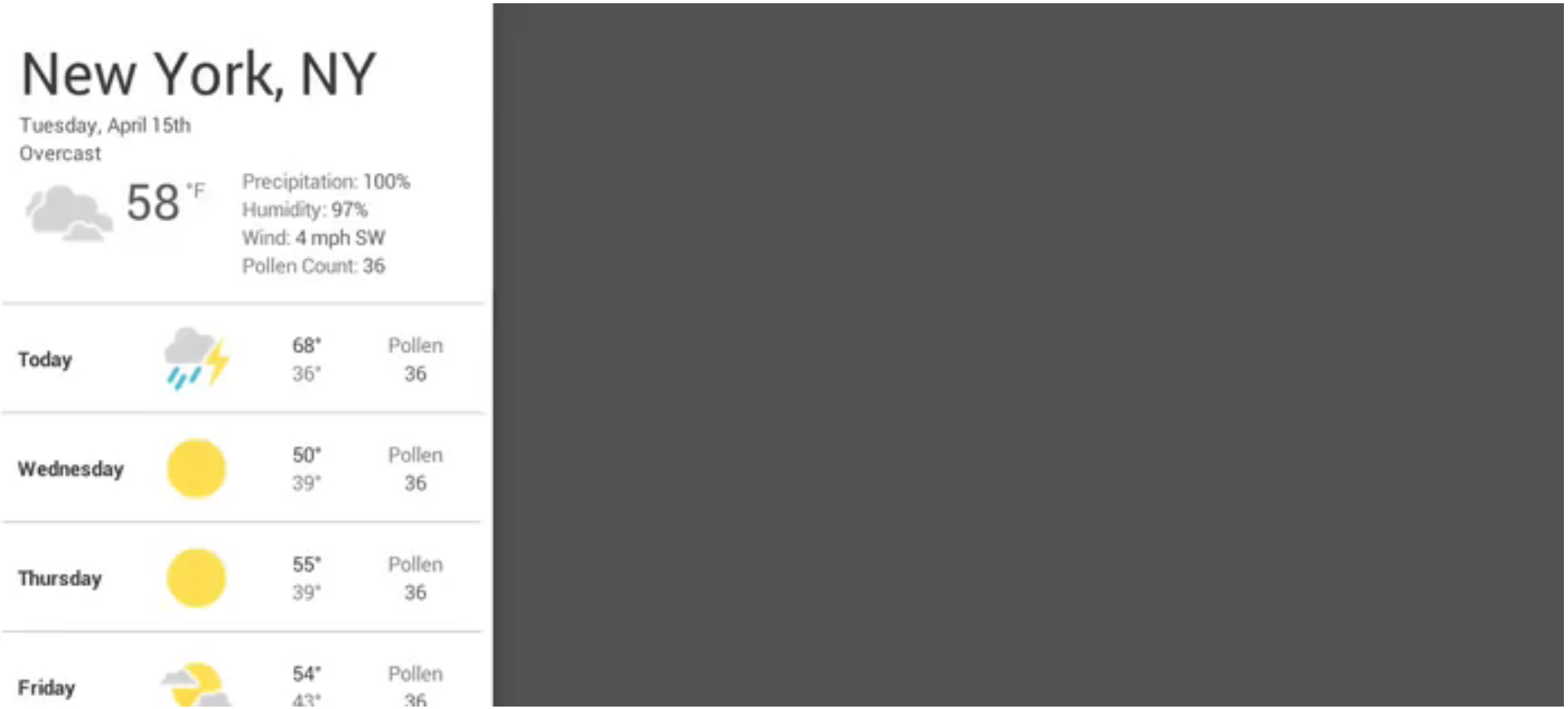
Smallest screen :
2 columns layout



Mobile:
1 column layout

Responsive Design Example

https://web.dev/responsive-web-design-basics/



More info

- https://www.w3schools.com/css/css_rwd_intro.asp
- https://web.dev/articles/responsive-web-design-basics
- https://www.simplilearn.com/tutorials/css-tutorial/css-responsive-design
- https://www.tutorialspoint.com/css/css_responsive.htm

What is The Viewport?

- The viewport is the user's visible area of a web page.
- The viewport varies with the device, and will be smaller on a cell phone than on a computer screen
- The viewport improves user experience on your site

What is The Viewport?

- To create a responsive website, add the following <meta> tag to all your web pages:

```
<meta name="viewport"
 content="width=device-width, initial-scale=1">
```

- **width=device-width:** sets the width of the page to follow the screen-width of the device

- **initial-scale=1.0 :** sets the initial zoom level when the page is first loaded by the browser.

Without the viewport meta tag

With the viewport meta tag

Media Queries

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What is a Media Query?

- **Media query** is a CSS technique introduced in CSS3
- It uses the **@media** rule to include a block of CSS properties only if a certain condition is true

Why should you use Media Queries?

- At first, the best solution seemed to be creating multiple stylesheets for different screen sizes and devices.
- Now, a single stylesheet can service a site on multiple devices, with responsive design. Less code to look after.
- Responsive design removes the need for serving up a separate stylesheet for each device, but it still uses the standard practice of detecting the user's device first.

Media Query Examples

```
@media only screen and (max-width: 600px) {
	body {
		background-color: lightblue;
	}
}

@media only screen and (min-width: 601px) and (max-width: 1200px) {
	body {
		background-color: pink;
	}
}

@media only screen and (min-width: 1201px)  {
	body {
		background-color: green;
	}
}
```

Orientation: Portrait / Landscape

- Media queries can also be used to change layout of a page depending on the orientation of the browser.

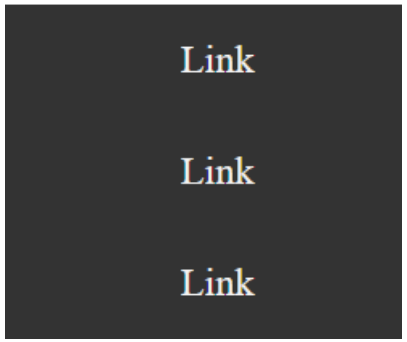
```
@media only screen and (orientation: landscape) {
  body {
    background-color: lightblue;
  }
}
```

practice with: media-queries-example.html

Media Queries For Menus

```
.topnav a {
   float: left;
   display: inline;
   text-align: center;
   text-decoration: none;
 }

 @media screen and (max-width: 600px) {
   .topnav a {
     float: none;
     width: 100%;
   }
 }
```

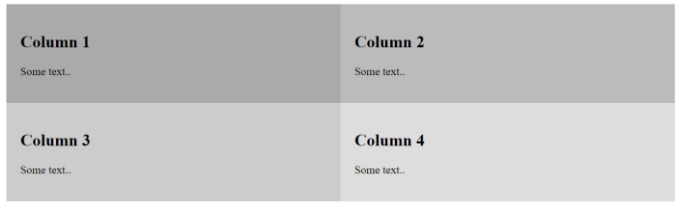


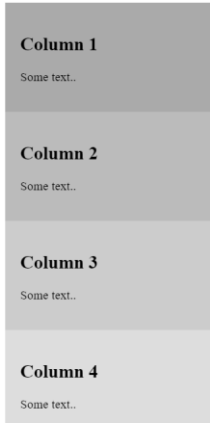
Media Queries For Columns

```
.column {
  float: left;
  width: 25%;
}

@media screen and (max-width: 992px) {
  .column {
    width: 50%;
  }
}

@media screen and (max-width: 600px) {
  .column {
    width: 100%;
  }
}
```





More Info

- https://developer.mozilla.org/en-US/docs/Web/CSS/Media_Queries/Using_media_queries

Responsive Web Design - Frameworks

- All popular CSS Frameworks offer responsive design.
  - Bootstrap
  - Pure CSS
  - Skeleton
  - SemanticUI
  - Foundation
  - W3.CSS
  - Tailwind

What is Bootstrap?

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What is Bootstrap?

- Bootstrap is the world's most popular framework for building responsive, mobile-first sites
  - free front-end framework for faster and easier web development
- Bootstrap includes HTML and CSS based design templates
  - forms, buttons, tables, navigation, modals, image carousels and many other, as well as optional JavaScript plugins

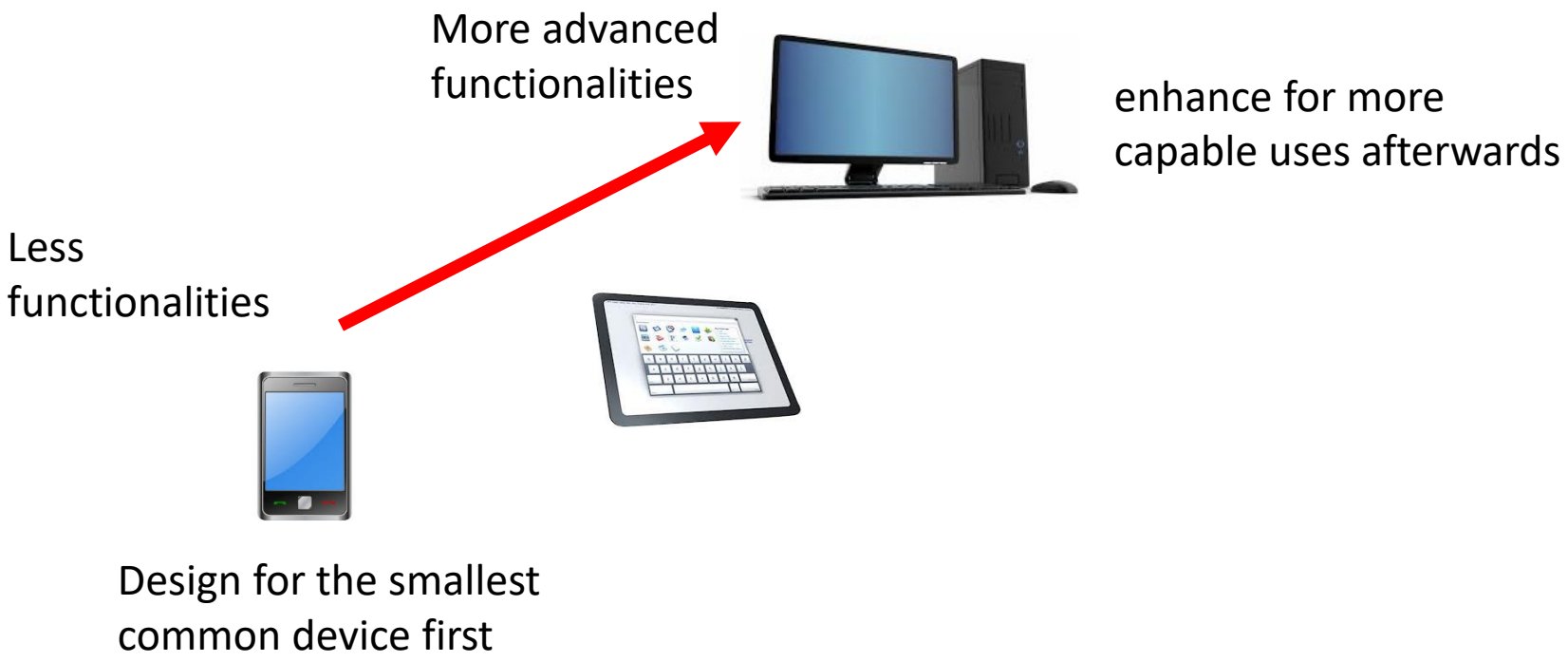
http://getbootstrap.com/

Bootstrap History

- Bootstrap was developed by Mark Otto and Jacob Thornton at Twitter
- Released as an open-source product in August 2011 on GitHub.
- In June 2014 Bootstrap was the No.1 project on GitHub!

What's Mobile First?

- A "mobile-first" approach involves designing a desktop site starting with the mobile version, which is then adapted to larger screens



Why Bootstrap?

- It makes life easier
- Easy to use web elements
- Relatively easy to learn
- Compatible with all modern browsers
- Wide range of themes/templates
- Quick responsive websites
- Great documentation
- Open source

More Info

- https://getbootstrap.com/docs/5.3/getting-started/introduction
- https://www.w3schools.com/bootstrap5/index.php

Get Started with Bootstrap

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Get started with Bootstrap

- Quick Start Documentation:
  - https://getbootstrap.com/docs/5.3/getting-started/introduction/
- Add Bootstrap to your project
  - Use CDN
    - https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css
  - Use Package manager
    - npm install bootstrap
  - Download
    - https://getbootstrap.com/docs/5.3/getting-started/download/

HTML 5 doctype

- Bootstrap uses HTML elements and CSS properties that require the HTML5 doctype.
- Always include the HTML5 doctype

```
<!DOCTYPE html>
<html lang="en">

</html>
```

Responsive meta tag

- Add responsive **viewport** meta tag to your <head> section

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

Quick Start

- https://getbootstrap.com/docs/5.3/getting-started/introduction/#quick-start
- Create a new index.html file in your project root.
- Include the <meta name="viewport"> tag as well for proper responsive behavior in mobile devices.
- Include Bootstrap's CSS and JS.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1"
    <title>Bootstrap demo</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bo
  </head>
  <body>
    <h1>Hello, world!</h1>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bo
  </body>
</html>
```

Layout

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Breakpoints

- Breakpoints are customizable widths that determine how your responsive layout behaves across device or viewport sizes in Bootstrap.
- Core concepts
  - Breakpoints are the building blocks of responsive design.
  - Use media queries to architect your CSS by breakpoint.
  - Mobile first, responsive design is the goal.

Available Breakpoints

Breakpoint	Class infix	Dimensions
X-Small	*None*	<576px
Small	sm	≥576px
Medium	md	≥768px
Large	lg	≥992px
Extra large	xl	≥1200px
Extra extra large	xxl	≥1400px

```
// Extra small devices (portrait phones, less than 576px)
// No media query for `xs` since this is the default in
// Bootstrap

// Small devices (landscape phones, 576px and up)
@media (min-width: 576px) { ... }

// Medium devices (tablets, 768px and up)
@media (min-width: 768px) { ... }

// Large devices (desktops, 992px and up)
@media (min-width: 992px) { ... }

// Extra large devices (large desktops, 1200px and up)
@media (min-width: 1200px) { ... }

// XXL devices (large desktops, 1400px and up)
@media (min-width: 1400px) { ... }
```

These breakpoints are customizable

Containers

- Containers are the most basic layout element in Bootstrap and are **required when using our default grid system**.
- Containers are a fundamental building block of Bootstrap that contain, pad, and align your content within a given device or viewport.
- Bootstrap comes with three different containers:
  - .container
  - .container-fluid
  - .container-{breakpoint}

Container Classes…

- .container:
  - responsive, fixed-width container, meaning its max-width changes at each breakpoint.
  - Default container

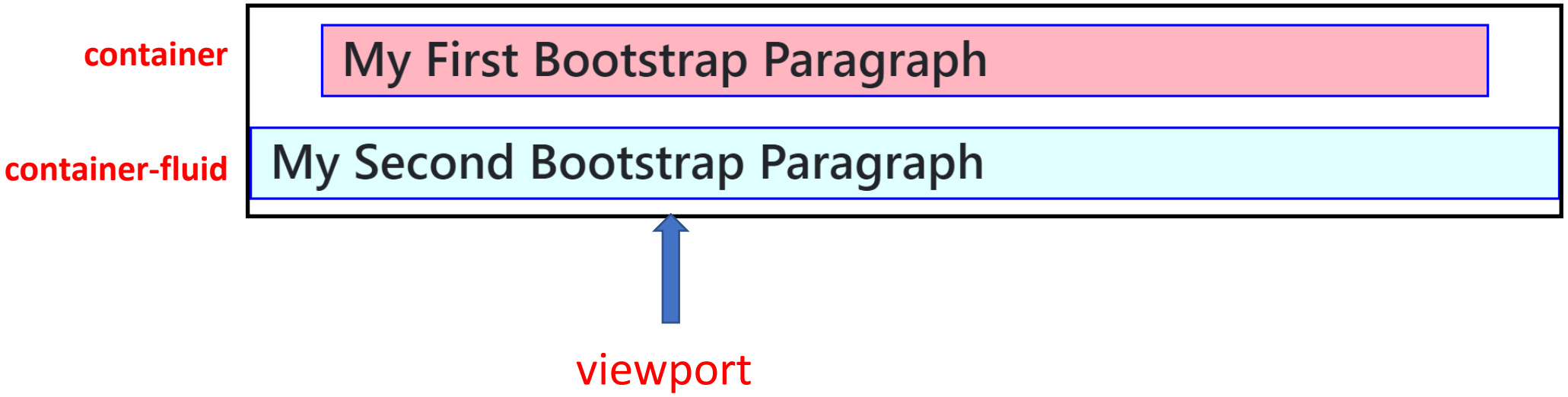
	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	X-Large ≥1200px	XX-Large ≥1400px
`.container`	100%	540px	720px	960px	1140px	1320px

- .container-fluid:
  - full width container, spanning the entire width of the viewport.

...Container Classes

```
<div class="container">
  <h1>My First Bootstrap Paragraph</h1>
</div>
```

```
<div class="container-fluid">
  <h1>My Second Bootstrap Paragraph</h1>
</div>
```



Practice with: bs-container-example.html

Container Classes => Responsive Classes

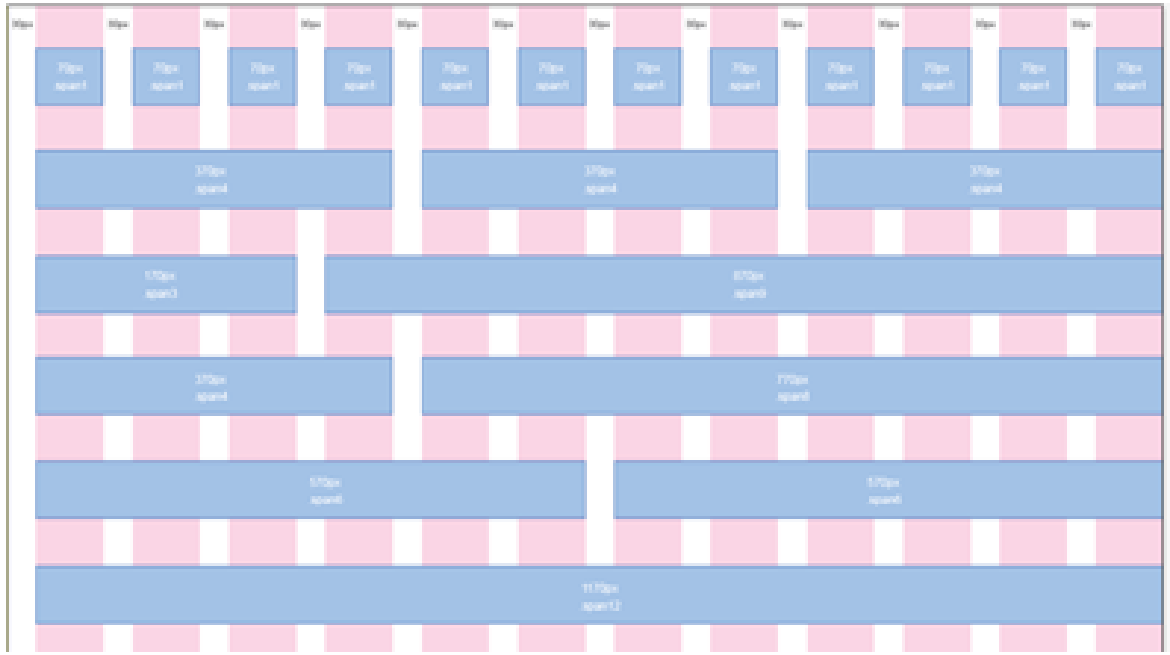
- They allow you to specify a class that is 100% wide until the specified breakpoint is reached, after which we apply max-widths for each of the higher breakpoints.

https://getbootstrap.com/docs/5.3/layout/containers/

Class	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	X-large ≥1200px	XX-Large ≥1400px
.container-sm	100%	540px	720px	960px	1140px	1320px
.container-md	100%	100%	720px	960px	1140px	1320px
.container-lg	100%	100%	100%	960px	1140px	1320px
.container-xl	100%	100%	100%	100%	1140px	1320px
.container-xxl	100%	100%	100%	100%	100%	100%

Grid system

- Bootstrap grid system is broken into 12 columns
  - to create wider columns, group the columns together
- Examples: https://getbootstrap.com/docs/5.3/layout/grid/



How it works?

- Bootstrap's grid system uses a series of containers, **rows**, and **columns** to layout and align content.

Column	Column	Column

```
<div class="container">
  <div class="row">
    <div class="col">
      Column
    </div>
    <div class="col">
      Column
    </div>
    <div class="col">
      Column
    </div>
  </div>
</div>
```

Grid System Rules

- Containers provide a means to center and horizontally pad your site's contents.
- Use *.container* for a responsive pixel width or *.container-fluid* for width: 100% across all viewport and device sizes.
- Rows are wrappers for columns.
- In a grid layout, content must be placed within columns and only columns may be immediate children of rows.
- For more rules: https://getbootstrap.com/docs/5.3/layout/grid/#how-it-works

Six responsive breakpoints

Breakpoint	Class infix	Dimensions
X-Small	*None*	<576px
Small	sm	≥576px
Medium	md	≥768px
Large	lg	≥992px
Extra large	xl	≥1200px
Extra extra large	xxl	≥1400px

- Breakpoints are based on **min-width** media queries, meaning they **affect that breakpoint and all those above** it (e.g., .col-sm-4 applies to sm, md, lg, xl, and xxl).
- This means you can control container and column sizing and behavior by each breakpoint.

Grid Examples: 3 equal columns…

- Get three equal-width columns **starting at desktops and scaling to large desktops.**
- On mobile devices, tablets and below, the columns will automatically stack.

```
<div class="container">
  <div class="row">
    <div class="col-md-4">
      First of three columns
    </div>
    <div class="col-md-4">
      Second of three columns
    </div>
    <div class="col-md-4">
      Third of three columns
    </div>
  </div>
</div>
```

Breakpoint	Min width
X-Small	<576px
Small(sm)	≥576px
Medium(md)	≥768px
Large(lg)	≥992px
Extra large(xl)	≥1200px
Extra extra large (xxl)	≥1400px

...Grid Examples: 3 equal columns

- Laptop L-1440px

First of three columns	Second of three columns	Third of three columns

- Laptop-1024px

First of three columns	Second of three columns	Third of three columns

- Tablet: 768px

First of three columns	Second of three columns	Third of three columns

- For mobile: 425px

First of three columns
Second of three columns
Third of three columns

- For mobile: 375px

First of three columns
Second of three columns
Third of three columns

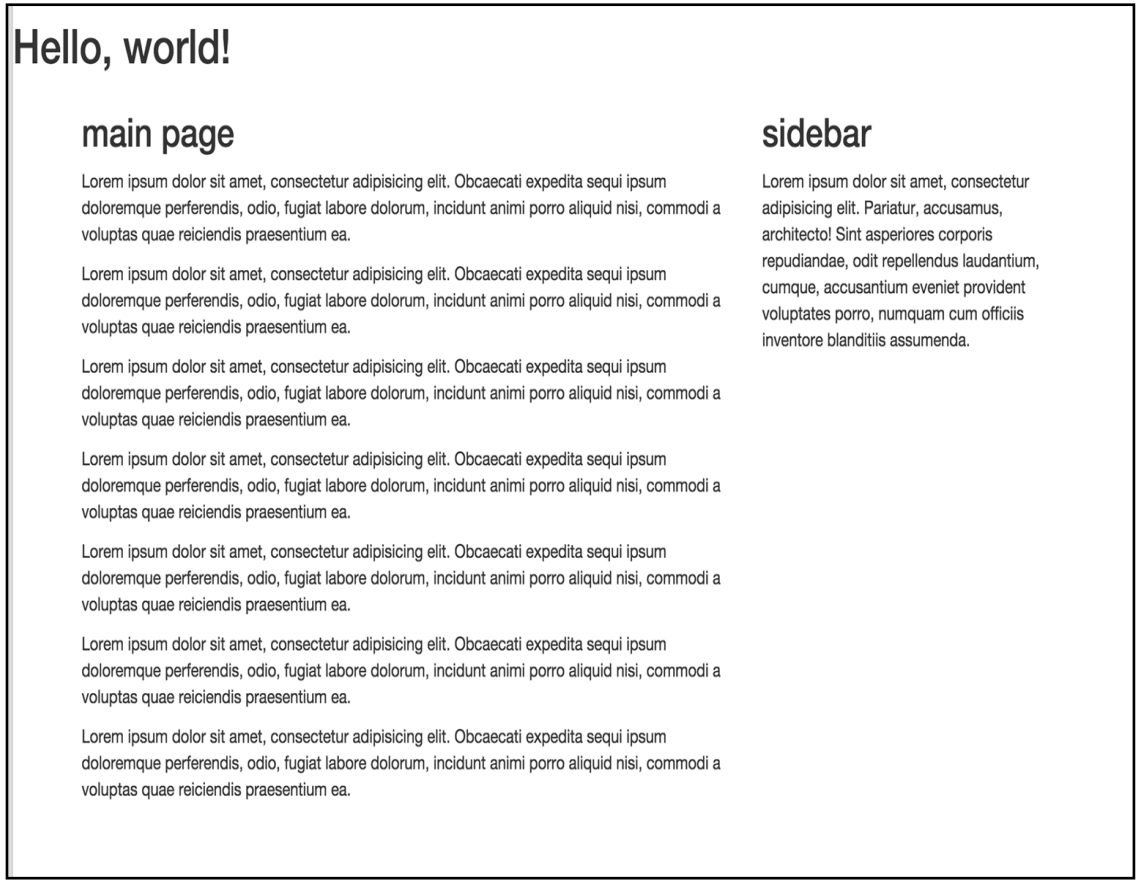
Grid Examples: Three unequal columns

- Get three columns starting at desktops and scaling to large desktops of various widths.

```
<div class="container">
  <div class="row">
    <div class="col-md-4">
      First of three columns
    </div>
    <div class="col-md-6">
      Second of three columns
    </div>
    <div class="col-md-2">
      Third of three columns
    </div>
  </div>
</div>
```

First of three columns	Second of three columns	Third of three columns

Grid Examples: Two unequal columns




```
<div class="container">
      <div class="row">
          <div class="col-sm-8">
              main page
          </div>
          <div class="col-sm-4">
              sidebar
          </div>
      </div>
 </div>
```

Mix and match

- Use a combination of different classes for each tier as needed

```
<div class="row">
          <div class="col-md-2 col-sm-4">TEXT</div>
          <div class="col-md-6 col-sm-4">TEXT</div>
          <div class="col-md-4 col-sm-4">TEXT</div>
</div>
```

TEXT	TEXT	TEXT

md and above devices

TEXT	TEXT	TEXT

Sm devices

Practice with:

- bs-grid-example1.html
- bs-grid-example2.html
- bs-grid-example3.html

Row columns…

- Use the responsive .row-cols-* classes to quickly set the number of columns that best render your content and layout.
- Whereas normal .col-* classes apply to the individual columns (e.g., .col-md-4), the row columns classes are set on the parent .row as a shortcut.

...Row columns

```
<div class="container">
    <div class="row row-cols-1 row-cols-sm-2 row-cols-lg-4">
        <div class="col">TEXT-1</div>
        <div class="col">TEXT-2</div>
        <div class="col">TEXT-3</div>
        <div class="col">TEXT-4</div>
        <div class="col">TEXT-5</div>
        <div class="col">TEXT-6</div>
    </div>
</div>
```

xs

TEXT-1
TEXT-2
TEXT-3
TEXT-4
TEXT-5
TEXT-6

sm, md

TEXT-1	TEXT-2
TEXT-3	TEXT-4
TEXT-5	TEXT-6

lg, xl, xxl

TEXT-1	TEXT-2	TEXT-3	TEXT-4
TEXT-5	TEXT-6		

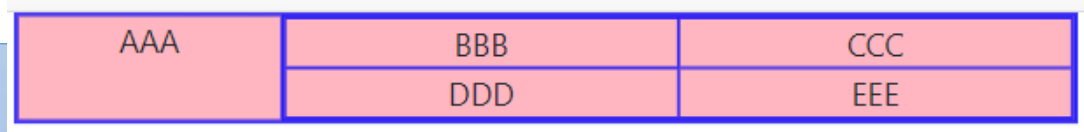
Nesting…

- To nest your content with the default grid, add a new .row and set of .col-sm-* columns within an existing .col-sm-* column.
- Nested rows should include a set of columns that add up to 12 or fewer

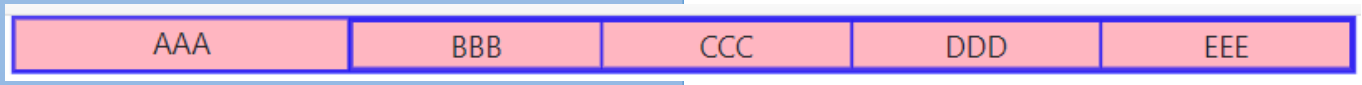
… Nesting

```
<div class="container">
    <div class="row">
        <div class="col-sm-3">AAA</div>
        <div class="col-sm-9">
            <div class="row">
                <div class="col-6 col-md-3">BBB</div>
                <div class="col-6 col-md-3">CCC</div>
                <div class="col-6 col-md-3">DDD</div>
                <div class="col-6 col-md-3">EEE</div>
            </div>
        </div>
    </div>
</div>
```

sm



md, lg, xl, xxl



Components

1. Responsive Web Design
2. Media Queries
3. What is Bootstrap?
4. Get Started with Bootstrap
5. Layout
6. Content & Components

Components

https://getbootstrap.com/docs/5.3/components/

- Accordion
- Alerts
- Badge
- Breadcrumb
- Buttons
- Button group
- Card
- Carousel
- Close button
- Collapse
- Dropdowns
- List group
- Modal
- Navs & tabs
- Navbar
- Offcanvas
- Pagination
- Placeholders
- Popovers
- Progress
- Scrollspy
- Spinners
- Toasts
- Tooltips

Alerts

- Provide feedback messages for typical user actions
- For proper styling, use one of the eight required contextual classes (e.g., .alert-success).

A simple primary alert—check it out!

A simple secondary alert—check it out!

A simple success alert—check it out!

A simple danger alert—check it out!

A simple warning alert—check it out!

```
<div class="alert alert-primary" role="alert">
  A simple primary alert—check it out!
</div>
<div class="alert alert-secondary" role="alert">
  A simple secondary alert—check it out!
</div>
<div class="alert alert-success" role="alert">
  A simple success alert—check it out!
</div>
<div class="alert alert-danger" role="alert">
  A simple danger alert—check it out!
</div>
<div class="alert alert-warning" role="alert">
  A simple warning alert—check it out!
</div>
. . .
```

Breadcrumb

- Indicate the current page's location within a navigational hierarchy

Home / Library / Data

```
<nav aria-label="breadcrumb">
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item"><a href="#">Library</a></li>
    <li class="breadcrumb-item active" aria-current="page">Data</li>
  </ol>
</nav>
```

Buttons

- Bootstrap includes several predefined button styles,

Primary Secondary Success Danger Warning Info Light Dark Link

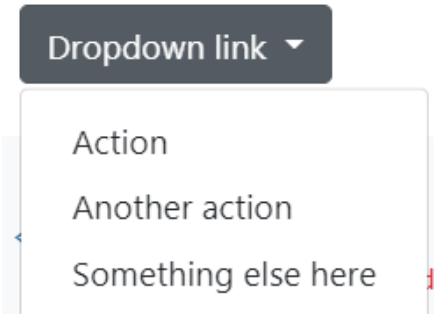
```
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-dark">Dark</button>
<button type="button" class="btn btn-link">Link</button>
```

Collapse

- Toggle the visibility of content
- The collapse JavaScript plugin is used to show and hide content.
- Go to page:
  - https://getbootstrap.com/docs/5.3/components/collapse/#example

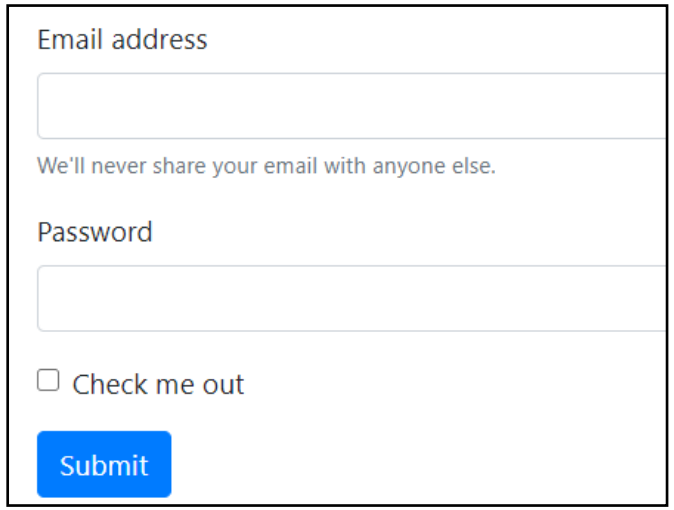
Dropdowns

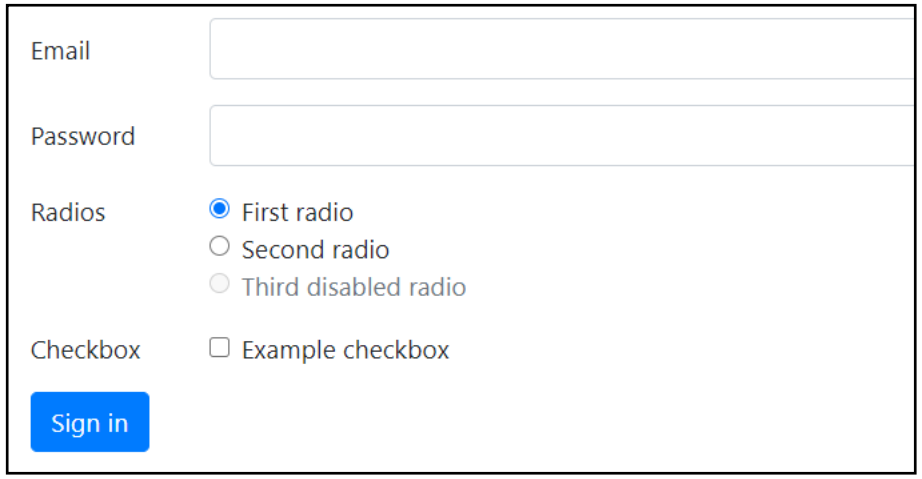
- Display lists
- Go to page:
  - https://getbootstrap.com/docs/5.3/components/dropdowns/#examples



Forms

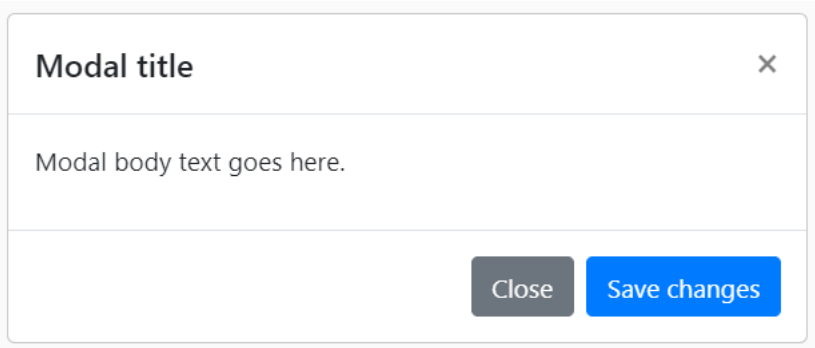
- Components for creating a wide variety of forms.
- Go to page:
  - https://getbootstrap.com/docs/5.3/forms/overview/





Modal

- Add dialogs to your site for lightboxes, user notifications, or completely custom content
- Go to page:
  - https://getbootstrap.com/docs/5.3/components/modal/



Navs

- Navigation components
- Go to page:
  - https://getbootstrap.com/docs/5.3/components/navs/

Active Link Link Disabled

```
<ul class="nav">
  <li class="nav-item">
    <a class="nav-link active" href="#">Active</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#" tabindex="-1" aria-disabled="true">Disabled</a>
  </li>
</ul>
```

For More Component Examples

- Go to page:
  - https://getbootstrap.com/docs/5.3/examples/

Responsive images

- Images in Bootstrap are made responsive with *.img-fluid. max-width: 100%; and height: auto;* are applied to the image so that it scales with the parent element.

```
<img src="..." class="img-fluid" alt="Responsive image">
```

Tables

- Go to page:
  - https://getbootstrap.com/docs/5.3/content/tables/#examples

#	First Name	Last Name	Username
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry	the Bird	@twitter

```
<table class="table">
...
</table>
```

#	First	Last	Handle
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry	the Bird	@twitter

```
<table class="table table-striped"> ...
</table>
```

Any Question?

Thank you for your attention

esma.meral@ozyegin.edu.tr